Now from Fig.5.7, it is clear that

Area of ΔABC

= Area of trapezium ABQP + Area of trapezium APRC − Area of trapezium $BQRC$.

You also know that, the area of trapezium

$$= \frac{1}{2} \times (\text{sum of parallel sides}) \times (\text{perpendicular distance between the parallel sides})$$

Therefore, Area of ΔABC

$$= \frac{1}{2}(BQ + AP)QP + \frac{1}{2}(AP + CR)PR - \frac{1}{2}(BQ + CR)QR$$

$$= \frac{1}{2}(y_2 + y_1)(x_1 - x_2) + \frac{1}{2}(y_1 + y_3)(x_3 - x_1) - \frac{1}{2}(y_2 + y_3)(x_3 - x_2)$$

$$= \frac{1}{2}\left\{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right\}$$

Thus, the area of ΔABC is the absolute value of the expression

$$= \frac{1}{2}\left\{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right\} \text{ sq.units.}$$

The vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ of ΔABC are said to be "taken in order" if A, B, C are taken in anticlockwise direction. If we do this, then area of ΔABC will never be negative.

Another form

The following pictorial representation helps us to write the above formula very easily.

$$\text{Area of } \Delta ABC = \frac{1}{2}\begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$$

$$= \frac{1}{2}\left\{(x_1 y_2 + x_2 y_3 + x_3 y_1) - (x_2 y_1 + x_3 y_2 + x_1 y_3)\right\} \text{ sq.units.}$$

Note

"As the area of a triangle can never be negative, we must take the absolute value, in case area happens to be negative".

Progress Check

The vertices of ΔPQR are $P(0, -4)$, $Q(3, 1)$ and $R(-8, 1)$

1. Draw ΔPQR on a graph paper.
2. Check if ΔPQR is equilateral.
3. Find the area of ΔPQR.
4. Find the coordinates of M, the mid-point of QP.
5. Find the coordinates of N, the mid-point of QR.
6. Find the area of ΔMPN.
7. What is the ratio between the areas of ΔMPN and ΔPQR?

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